SRIRAM VIVEK

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Projects

Double-Entry Ledger and Payment Engine | GitHub ↗

- Built a payment system on Kafka and PostgreSQL that takes requests immediately, completes them in the background through Stripe (test mode), and stays correct under retries and crashes
- Eliminated double charges on client retries, taking 1,000 concurrent identical requests from 40 payments to 1, by adding API-level idempotency keys
- Recovered stranded payments due to API crash by adding a transactional outbox between PostgreSQL and Kafka
- Ensured every payment is fully accounted for by recording equal debits and credits in a double-entry ledger that rejects unbalanced entries, verified by 10 pytest integration tests

Flight Booking Microservices System | GitHub ↗

- Designed a flight booking backend of 3 Python microservices (User, Search, Orchestrator) over gRPC
- Raised booking success from 49% to 91.5% when half of Search calls failed, using retries with exponential backoff, and cut calls to a fully down service from 300 to 11 per 100 requests with a circuit breaker
- Secured service-to-service calls with mutual TLS so only authenticated services can connect
- Added distributed tracing (OpenTelemetry, Jaeger) and metrics (Prometheus, Grafana) to debug slow requests

Agentic RAG System with Custom MCP Server | GitHub ↗

- Built a RAG agent on Qdrant answering questions from Anthropic API docs and declining questions they don't cover
- Grounded every answer in retrieved sources by limiting the agent to 4 retrieval tools on a custom MCP server
- Surfaced retrieved passages, similarity scores, and citations in a Next.js dashboard so users can verify answers
- Deployed with GitHub Actions CI/CD to Vercel and a Linux server behind Nginx

Experience

Galatea Associates LLC | Financial Software Engineer Intern — **June 2025 – August 2025**
Software firm building custom systems for Wall Street investment banks — Boston, Massachusetts

- Contributed to an LLM proof of concept for a financial services client that explains a portfolio optimizer's numeric trade suggestions in plain language
- Wrote 3 multi-table SQL queries in MS SQL Server that fed the LLM suggested trades, industry allocations before and after optimization, and a benchmark comparison, all adopted into the product
- Cut manual document searching for the team by building a RAG pipeline (LangChain, OpenAI embeddings, PGVector) over a 600-page optimizer specification that adds relevant behavior to the LLM prompt

Skills

Development Tools	Claude Code CLI, Codex, Cursor, Git
Programming Languages	Python, SQL
Databases	PostgreSQL, MS SQL Server, Qdrant
Backend & Distributed Systems	FastAPI, REST APIs, gRPC, Protocol Buffers, Apache Kafka, event-driven architecture, microservices, transactional outbox, dead-letter queue (DLQ), pytest
Infrastructure & Observability	AWS (S3, EC2, Lambda), Docker, CI/CD (GitHub Actions), Linux, Nginx, systemd, Prometheus, Grafana, Jaeger, OpenTelemetry, distributed tracing
Generative AI & LLMs	RAG, LangChain, Model Context Protocol (MCP)

Education

Stony Brook University — **August 2024 – May 2026**
Master of Science in Computer Science and Applied Mathematics (Data Science) — New York, United States
Courses: Data Structures and Algorithms, Computer Networks, Programming Abstractions, Theory of Computation, Data Management, Big Data Systems, Big Data Analysis, Probability, Statistical Learning, Statistical Computing

SSN College of Engineering — **November 2020 – May 2024**
Bachelor of Engineering in Electrical and Electronics Engineering — Chennai, India